Kristófer D. Kristjánsson

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SUMMARY

Passionate M.Sc. Autonomous Systems student with hands-on experience in robotics and development of computer vision applications and sensor integration. Skilled in Python, real-time imaging pipelines, and computer vision with a strong interest in developing autonomy for mobile or stationary systems.

EDUCATION

Denmark Technical University — **Lyngby, Denmark**
M.Sc. Autonomous Systems — Graduation Date: 2027

University of Reykjavik — **Reykjavik, Iceland**
B.Sc. Mechatronics Engineering — Graduation Date: 2024

RELEVANT WORK EXPERIENCE

RiACT — **Copenhagen, DK**
Junior Robotics Engineer — *02/2026-Current*
• Built Python and Linux automation tools for inside use, simplifying workflows in an agile team.
• Designed and implemented a Python-based API client for camera server integration, enabling accurate image and detection retrieval.
• Developed hand-eye calibration tool for 3rd party camera systems, including driver implementation, testing, and optical calibration.

Verra Mobility — **Arizona, USA**
Hardware Engineer — *03/2024-07/2025*
• Troubleshooted and resolved traffic camera issues related to both software and hardware reasons.
• Designed and performed quality control processes intended to decrease electrical and software malfunctions.
• Led projects in collaboration with multiple departments and companies on joint projects.

Arizona State University — **Arizona, USA**
Summer Undergraduate Research Internship — *06/2023-08/2023*
• Developed automated microchip measurement system using PLCs, DAQ, and motion control, reducing test time from hours to under 60 seconds.
• Awarded the **SURI 2023 Fellowship** in recognition of exceptional research contributions.

RELEVANT PROJECTS

Greenhouse Tomato Detection System - YOLOv8 Perception Pipeline

• Built an end to end object detection pipeline for autonomous greenhouse harvesting, from data collection and manual annotation through training and evaluation.
• Fine tuned a YOLOv8n model achieving a 93% precision and 90-95% mAP50, handling real world challenges including partial occlusion, clustered growth and variable lighting.

Vision Based Process Monitoring for Automated Synthesis

• Built real time multi object tracking pipeline (YOLOv8n detection + DeepSORT, OpenCV/NumPy) monitoring process stages via per-object feature extraction (HSV, texture, and temporal analysis) for anomaly detection.
• Designed label free anomaly detection method using batch statistics to flag failures without supervised training data addressing the low-sample constraint.

RELEVANT SKILLS

Software: C++, Python, Ubuntu Linux, Jupyter Notebook, PyTorch, OpenCV, Numpy, Git, ROS2, Onshape
Hardware: Oscilloscope, Multimeter, ESP32/Raspberry PI, Soldering